

■ 1.4.7 Advanced Topic: Putting Expressions into Different Forms

Complicated algebraic expressions can usually be written in many different ways. *Mathematica* provides a variety of functions for converting expressions from one form to another.

In most applications, the commonest of these functions are **Expand**, **Factor** and **Simplify**. However, particularly when you have “rational” expressions that contain quotients, there are other functions that you may need to use.

Expand [<i>expr</i>]	multiply out products and powers
ExpandAll [<i>expr</i>]	apply Expand everywhere
Factor [<i>expr</i>]	reduce to a product of factors
FactorTerms [<i>expr</i>]	pull out common factors from each term
Together [<i>expr</i>]	put all terms over a common denominator
Apart [<i>expr</i>]	separate into terms with simple denominators
Cancel [<i>expr</i>]	cancel common factors between numerators and denominators
Collect [<i>expr</i> , <i>x</i>]	group together powers of <i>x</i>
Simplify [<i>expr</i>]	try a sequence of algebraic transformations and give the smallest form of <i>expr</i> found

Functions for transforming algebraic expressions.

Here is a rational expression that can be written in many different forms.

$$\begin{aligned} \text{In}[1] := e &= (x - 1)^2 (2 + x) / ((1 + x) (x - 3)^2) \\ \text{Out}[1] &= \frac{(-1 + x)^2 (2 + x)}{(-3 + x)^2 (1 + x)} \end{aligned}$$

Expand expands out the numerator, but leaves the denominator in factored form.

$$\begin{aligned} \text{In}[2] := \text{Expand}[e] \\ \text{Out}[2] &= \frac{2}{(-3 + x)^2 (1 + x)} - \frac{3x}{(-3 + x)^2 (1 + x)} + \frac{x^3}{(-3 + x)^2 (1 + x)} \end{aligned}$$

ExpandAll expands out everything, including the denominator.

$$\begin{aligned} \text{In}[3] &:= \text{ExpandAll}[e] \\ \text{Out}[3] &= \frac{2}{9 + 3x - 5x^2 + x^3} - \frac{3x}{9 + 3x - 5x^2 + x^3} + \\ &\quad \frac{x^3}{9 + 3x - 5x^2 + x^3} \end{aligned}$$

Together collects all the terms together over a common denominator.

$$\begin{aligned} \text{In}[4] &:= \text{Together}[\%] \\ \text{Out}[4] &= \frac{2 - 3x + x^3}{9 + 3x - 5x^2 + x^3} \end{aligned}$$

Apart breaks the expression apart, into terms with simple denominators.

$$\begin{aligned} \text{In}[5] &:= \text{Apart}[\%] \\ \text{Out}[5] &= 1 + \frac{5}{(-3 + x)^2} + \frac{19}{4(-3 + x)} + \frac{1}{4(1 + x)} \end{aligned}$$

Factor factors everything, in this case reproducing the original form.

$$\begin{aligned} \text{In}[6] &:= \text{Factor}[\%] \\ \text{Out}[6] &= \frac{(-1 + x)^2 (2 + x)}{(-3 + x)^2 (1 + x)} \end{aligned}$$

According to **Simplify**, the original form is the simplest way to write the expression.

$$\begin{aligned} \text{In}[7] &:= \text{Simplify}[e] \\ \text{Out}[7] &= \frac{(-1 + x)^2 (2 + x)}{(-3 + x)^2 (1 + x)} \end{aligned}$$

Getting algebraic expressions into the form you want is something of an art. In most cases, the best thing to do is to “experiment”, trying different transformations until you get what you want.

When you have an expression in a single variable, you can choose to write it as a sum of terms, a product, and so on. If you have an expression with several variables, there is an even wider selection of possible forms. You can, for example, choose to group terms in the expression so that one or another of the variables is “dominant”.

Here is an algebraic expression in two variables.

```
In[8]:= v = Expand[(3 + 2x + y)^3]
```

```
Out[8]= 27 + 54 x + 36 x^2 + 8 x^3 + 27 y + 36 x y +
        12 x^2 y + 9 y^2 + 6 x y^2 + y^3
```

This groups together terms in v that involve the same power of x .

```
In[9]:= Collect[v, x]
```

```
Out[9]= 27 + 8 x^3 + 27 y + 9 y^2 + y^3 + x^2 (36 + 12 y) +
        x (54 + 36 y + 6 y^2)
```

This groups together powers of y .

```
In[10]:= Collect[v, y]
```

```
Out[10]= 27 + 54 x + 36 x^2 + 8 x^3 + y^3 + y^2 (9 + 6 x) +
        y (27 + 36 x + 12 x^2)
```

Even when one restricts oneself to algebraic expressions, there are many different ways to write any particular expression. If one goes beyond purely algebraic expressions, and includes, for example, higher mathematical functions, the variety of possible forms becomes still greater. As a result, it is totally infeasible to have a specific function in *Mathematica* to produce each possible form. Instead, as described in Chapter 2, *Mathematica* allows you to set up arbitrary transformation rules to convert between different forms.